3. Part A — HTTP Methods & Message

A1 — Audit verbs

Check every endpoint from Assignment 4. GET reads, POST creates/actions, PUT replaces, PATCH modifies, DELETE removes. Avoid action verbs such as /cancelOrder when a POST sub-resource is appropriate.

A2 — Non-CRUD actions

Use POST to a sub-resource: POST /api/executions/101/cancel, POST /api/executions/101/reset, and POST /api/algorithms/1/execute.

A3 — Safe & idempotent

Endpoint type	Safe?	Idempotent?	Retry note
GET algorithm/read	Yes	Yes	Naturally retry-safe
POST execute	No	No	Use Idempotency-Key
PUT execution	No	Yes	Use If-Match for concurrency
PATCH execution	No	Depends	Use If-Match where appropriate
DELETE history	No	Yes	Repeated delete has same final state
POST cancel/reset	No	Depends	Use Idempotency-Key if duplicate action is harmful

A4 — Query parameters

```
GET /api/algorithms?category=sorting&sort=name&page=0&size=10
```

Keep the list endpoint a pure GET/read.

A5 — OPTIONS + Allow

Implement OPTIONS for one resource and return an Allow header. If a constrained client needs a documented fallback, accept X-HTTP-Method-Override for PUT/DELETE.

A6 — Full exchange

In NOTES.md include one complete request and response: request line with HTTP version, Host, headers and JSON body; then status line, headers and body.